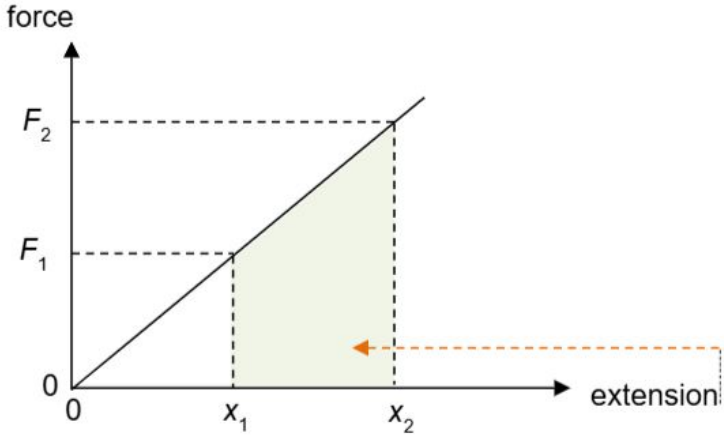
3 The force-extension graph of a spring is as shown.



What is the change in elastic potential energy when the extension of the spring is increased from x_1 to x_2?

A $F_2x_2 - F_1x_1$

B $\frac{1}{2}(F_2x_2 - F_1x_1)$

C $F_2x_2^2 - F_1x_1^2$

D $\frac{1}{2}(F_2x_2^2 - F_1x_1^2)$

Ans: B

The area under the force-extension graph gives the elastic potential energy stored in the spring.

As extension is increased from x_1 to x_2, the area changes from $\frac{1}{2}F_1x_1$ to $\frac{1}{2}F_2x_2$.

Hence the change in EPE = $\frac{1}{2}(F_2x_2 - F_1x_1)$.